

Jaehyun Jeon

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🌐 [Homepage](#) in [LinkedIn](#) [Github](#)

Experience

NC AI

Research Engineer (DialogTech Team)

Jan. 2026 – Current

- Developed and evaluated domain-specialized AI agents for game customer support.
- Built an internal eval and trace visualization platform, designed checklist-based eval metrics, and a configurable experiment system, iterating over tool selection and context compression strategies, achieving 30pp quality improvement.
- Collaborated with game customer support agents to define output schema and field priorities for existing specialized workflows.

LG AI Research

Research Engineer Internship (Language Lab)

Sep. 2024 – Dec. 2025

- Built ChatEXAONE Extended Thinking, an agentic feature with integrated web search and patent-specialized workflows tailored to legal document structure, trained on QA-derived trajectory data.
- Served as internal design manager, ensuring production-ready service behavior working between model development and UX, including streaming integration and citation rendering.

MIR Lab, Yonsei University (Advisor: Youngjae Yu)

Graduate Researcher

Mar. 2024 – Aug. 2025

- Conducted multimodal AI research/projects, focusing on real-world problem solving via model evaluation, alignment, and RAG

Research Internship

Mar. 2023 – Feb. 2024

- Assisted in multimodal AI projects, gaining practical skills in model training/inference at code level and performance analysis

Data Science Lab (DSL), Yonsei University (Student academic society)

Vice President

Jul. 2022 – Dec. 2022

- Society operations: recruitment, networking, session management, project acquisition & progress tracking
- Managed three industry colab projects: contracts and overall project lead (Freewheelin, Pozalabs, VisualCamp)

Acting Member

Jan. 2022 – Dec. 2022

- Led sessions/study groups on ML topics (Generative models, NLP, MLOps, etc), and contributed to industry/internal projects

Education

Yonsei University

M.S. in Computer Science (GPA: 4.24/4.3, Advisor: Youngjae Yu)

Mar. 2024 – Feb. 2026

B.S. in Computer Science (GPA: 4.26/4.3, Summa Cum Laude, Top 1/106)

Mar. 2018 – Feb. 2024

Publication (◦ Subject / ◦ Contribution)

🎓 [Google Scholar](#)

- [1] **Do MLLMs Capture How Interfaces Guide User Behavior? A Benchmark for Multimodal UI/UX Design Understanding (ACL 2026 Main)** [pdf](#) [code](#)
Jaehyun Jeon, Min Soo Kim, Jang Han Yoon, Sumin Shim, Yejin Choi, Hanbin Kim, Dae Hyun Kim[†], Youngjae Yu[†]
 - Proposed a multimodal LLM evaluation task for understanding and reasoning about human behavior in UI/UX design
 - Built a benchmark using real-world A/B-tested UI pairs with rationales, revealing significant MLLM limitations including near-random selection performance, and outlined future directions for behavior-aware UI generation
- [2] **A11yn: Aligning LLMs for Accessible Web UI Code Generation (Under review)** [pdf](#)
Janghan Yoon, Jaegwan Cho, Junhyeok Kim, Jiwan Chung, **Jaehyun Jeon**, Seungwon Lim, Youngjae Yu
 - Developed a GRPO-based framework to align code-generating LLMs with web UI accessibility standards
 - Built multi-GPU training pipeline with axe-core-based accessibility compliance reward function design and synthetic datasets
- [3] **Zero-shot Multimodal Document Retrieval via Cross-modal Question Generation (EMNLP 2025 Main)** [pdf](#)
Yejin Choi, Jaewoo Park, Janghan Yoon, Saejin Kim, **Jaehyun Jeon**, Youngjae Yu
 - Developed a multimodal retrieval framework achieving SOTA performance on out-of-distribution benchmarks
 - Proposed preQ generation method and demonstrated improved recall on multilingual Chinese and Korean RAG benchmarks

- [4] **Can Visual Language Models Resolve Textual Ambiguity with Visual Cues? Let Visual Puns Tell You!** (*EMNLP 2024 Main*) [pdf](#) [↗](#)
 Jiwan Chung, Seungwon Lim, Jaehyun Jeon, Seungbeen Lee, Youngjae Yu
- Introduced a benchmark evaluating multimodal literacy in models using images for pun understanding and word disambiguation
 - Developed multimodal pun benchmark through image generation/retrieval and conducted model performance evaluations

Project (◦ Subject / ◦ Contribution)

- APEX: AI-Powered Patent Specification Drafting Platform via Multi-Agent Pipeline** Apr. 2026 - Current
- Built a platform for automated patent specification drafting using a multi-agent pipeline with parallel subagents across claim drafting, drawing labeling, and iterative spec generation with escalation logic.
 - Led full-stack and model development, and collaborated iteratively with patent attorneys to define domain-specific rubrics as shared evaluation criteria for human review and model self-assessment loops.
- [Yanolja] Automated Accommodation Image Assessment Based on User Preferences** (*Lead*) Jun. 2025 - Aug. 2025
- Developed an end-to-end framework for assessing accommodation images based on human-perceived attractiveness
 - Built assessment framework identifying informativeness and aesthetics as crucial aspects, created manually labeled datasets, trained and evaluated VLMs, and proposed image sorting strategies to enhance user experience
- [LG Electronics] webOS Platform Development Support Multimodal Chatbot** May. 2024 - Dec. 2024
- Built RAG-based multimodal chatbot with webOS knowledge base, automating data preprocessing to answering pipeline
 - Designed document preprocessing pipeline with markdown conversion, built curated benchmark for performance analysis, implemented reranking to improve in answer quality, and delivered API for usability
- AkaLlama: Korean Specialized 70B LLM Construction** [link](#) [↗](#) May. 2024 - Dec. 2024
- Released a 70B LLaMA-based LLM full fine-tuned with curated data to be specialized in Korean
 - Curated Korean training dataset gathering open-source instruction-following samples through quality analysis and deduplication, and evaluated performance on major Korean benchmarks showing results comparable to leading models
- Visual Distractor: Vision-Language Models Fooled by Images** (*Lead*) Sep. 2023 - Dec. 2023
- Proposed a benchmark measuring VLM hallucination vulnerabilities from visual interference in text question answering
 - Built datasets by generating diverse questions on anomalous synthetic images and evaluated VLMs for incorrect responses
- User-interactive Image Captioning with Constrained Decoding** (*Lead*) [code](#) [↗](#) Mar. 2023 - Jun. 2023
- Developed an interactive image captioning system enabling users to fix specific words in the generated captions
 - Implemented constrained decoding for VLMs by modifying Transformers code to enforce fixed words during generation
- [Freewheelin] Establishment of Student Performance Indicators in Service** (*Lead*) Oct. 2022 - Dec. 2022
- Identified students' actual math skill levels to highlight the value of the company's service
 - Developed an embedding-based model predicting student math performance, enabling personalized study recommendations

Awards and Honors

- Scholarship**
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| Yonsei-to-Yonsei (Y2Y) Scholarship (Full, merit-based) | Mar. 2024 – Feb. 2026 |
| National Science & Technology Scholarship (Full, merit-based) | Feb. 2022 - Feb. 2024 |
| Academic Excellence Scholarship (Full, merit-based) | Sep. 2018 - Feb. 2022 |
- Awards**
- | | |
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| Software Capstone Design Project, Excellence Award (Fall Semester) | Dec. 2023 |
| Software Capstone Design Project, Excellence Award (Spring Semester) | Jun. 2023 |
| KPMG Ideathon Challenge, 3rd Place (Subj: 'Shopping Service with Carbon Emission Prediction') | Feb. 2022 |
| ICPC Sinchon Summer Algorithm Camp Contest, 2nd Place | Aug. 2021 |

Teaching

- Teaching Assistant**
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| CSI4101, 4102: Software Capstone Design | Sep. 2024 – Dec. 2024 |
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- Mentoring**
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| YDMC (Guiding high school students in career exploration) | Sep. 2023 – Jun. 2024 |
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Skills

Programming

[Language] Python, SQL, C++, JavaScript

[ML/DL & AI] PyTorch, Transformers (Huggingface), vLLM, LangChain

[Tools & Platforms] Linux, AWS, Docker, Git, WandB, MLflow, Figma

Language

Korean (Native), English (Fluent, 1.5 years working with US Army, TOEIC 950, OPIc IH)